Satyam Awasthi

Researcher & Software Engineer

satyam.aw@yahoo.com — satyam-aw.github.io — satyam-aw — 0000-0002-6095-4303 — LinkedIn

Professional Summary

Computer Science researcher focused on **Autonomous Systems**, **Spatial Perception**, and **Model-Based and Learning-Based Control**. My research explores how perception, predictive modeling, and planning can be integrated with control frameworks to enable robust long-horizon autonomy in robotic systems operating under uncertainty.

Research Interests

Autonomous Systems, Spatial Perception, Model-Based and Learning-Based Control, and Robot Planning, with emphasis on long-horizon decision-making under uncertainty in real-world robotic environments.

Education

University of California, Santa Barbara (UCSB) — Santa Barbara, CA
M.S. in Computer Science (GPA: 3.91/4.00) — Sep 2021 – Jun 2023

Indian Institute of Technology, Kharagpur (IIT Kharagpur) — Kharagpur, India
B.Tech. in Electrical Engineering, Minor in Computer Science (GPA: 8.98/10.0) — Jul 2016 – Jul 2020

Research Experience

Four Eyes Laboratory, University of California, Santa Barbara — Sep 2021 – Jun 2023
Graduate Researcher — Advisor: Prof. Tobias Höllerer, Co-Advisor: Prof. Michael Beyeler

- Developed **EyeTTS**, an open-source framework for evaluating and calibrating eye tracking during mixed-reality locomotion, resulting in first-author publications at IEEE ISMAR Adjunct and IEEE VR Workshops.
- Designed experimental protocols and software pipelines to quantify gaze-tracking drift during active user motion.
- Curated a multi-device eye-tracking dataset and conducted user studies across diverse locomotion layouts.
- Published first-author poster papers at IEEE VR and IEEE ISMAR Adjunct venues.

Autonomous Ground Vehicle & Swarm Robotics Group, IIT Kharagpur — Jan 2016 – Dec 2018
Undergraduate Researcher

- Contributed to the development of **Eklavya 6.0**, the IIT Kharagpur autonomous ground vehicle that finished **Runner-Up** at the **Intelligent Ground Vehicle Competition (IGVC) 2018**. Competition Video · Team AGV
- Developed a hardware-agnostic navigation stack integrating EKF-based localization from LiDAR, IMU, and GPS streams.
- Implemented visual obstacle segmentation using a custom U-Net architecture and hybrid motion planning using A*/Dijkstra and Dynamic Window Approach.
- Designed decentralized multi-agent coordination using relative-pose estimation over an ad-hoc mesh network.

Publications

Peer-Reviewed Conference Papers

SmartWatch for Wall Writing: Real-time Transcription of Wall Writing from Inertial Sensing
Snigdha Das, **Satyam Awasthi**, Abdul Shamnar P, Pradipta De, Sandip Chakraborty, and Bivas Mitra.
COMSNETS, 2022. DOI: 10.1109/COMSNETS53615.2022.9668531

Workshop & Poster Publications

Eye Tracking Performance in Mobile Mixed Reality
Satyam Awasthi, Vivian Ross, Sydney Lim, Michael Beyeler, and Tobias H"ollerer.
IEEE VRW, 2024. DOI: 10.1109/VRW62533.2024.00321

EyeTTS: Evaluating and Calibrating Eye Tracking for Mixed-Reality Locomotion
Satyam Awasthi, Vivian Ross, Michael Beyeler, and Tobias H"ollerer.
ISMAR Adjunct, 2023. DOI: 10.1109/ISMAR-Adjunct60411.2023.00104

Research Artifacts

EyeTTS Calibration Framework 2022–2024
`Calibration Framework` · `User Study Framework`

- Developed an extensible framework for evaluating and calibrating eye-tracking systems during mixed-reality locomotion.
- Implemented device-agnostic benchmarking pipelines and automated drift-correction procedures.
- Released reproducible software infrastructure supporting eye-tracking research.

EyeTTS Mixed-Reality Eye Tracking Dataset 2022–2024
`Dataset` · `User Study Videos`

- Curated a multi-device mixed-reality locomotion dataset for calibration and performance evaluation.
- Collected gaze trajectories, calibration measurements, and locomotion metadata across diverse MR layouts.
- Supported experimental validation reported in ISMAR and VRW publications.

Selected Projects

JitterNot Jan 2022 – Mar 2022
Learning-Based Model Predictive Control for Adaptive Video Streaming (UCSB)

- Developed an LSTM-based predictor for network throughput enabling closed-loop control under non-stationary dynamics.
- Designed a model predictive control (MPC) framework using receding horizon optimization for adaptive decision-making over bandwidth constraints.

IGVC Lane Detection & Navigation June 2018
ROS-Based Lane Perception Pipeline for Autonomous Ground Vehicles (IIT Kharagpur)

- Developed a ROS-integrated lane perception pipeline using OpenCV, bird's-eye-view projection, CUDA/GPU-accelerated SLIC-style superpixel segmentation, and parabolic lane fitting for real-time autonomous navigation.
- Rejected obstacle and non-lane regions using contour-based ROI extraction, HOG descriptors, and SVM classification, then published local waypoint goals and segmented top-view lane masks for ROS navigation.

In Motion — HAR using DeepConvLSTM Nov 2019
Embedded Sensor Inference System for Temporal Activity Recognition (IIT Kharagpur)

- Designed a streaming inference pipeline for high-frequency IMU signals enabling real-time temporal classification.
- Implemented a DeepConvLSTM architecture combining convolutional feature extraction with temporal sequence modeling for activity recognition.

Battleship Solitaire Solver Apr 2022 – Jun 2022
Solver-Aided Constraint Synthesis for Structured Search Problems (UCSB)

- Formulated Battleship Solitaire as a symbolic constraint-satisfaction problem in Rosette/Racket, encoding grid cells, ship placements, row/column tallies, adjacency rules, and revealed-cell constraints.
- Used Rosette's solver-aided execution to synthesize valid board configurations satisfying all puzzle constraints.

Industry Experience

Intuit Inc Mountain View, CA · Mar 2024 – Jan 2026
Software Engineer 2

- Built dual-stage **agentic decision pipelines** using tool-schema routing for real-time intent classification and sequential state tracking.
- Developed a decoupled **generative UI engine** leveraging RAG and structured LLMs to optimize deterministic component layouts.

Yahoo Inc Mountain View, CA · Apr 2023 – Nov 2023
Software Dev Engineer II (IC3)

- Developed **client-side rule engines** for asynchronous receipt payloads, executing deterministic filtering for **real-time decision support**.

Coupang Global LLC Mountain View, CA · Jun 2022 – Sep 2022
Software Engineering Intern

- Engineered **context-aware heuristics** to prune multi-dimensional search spaces based on active user state vectors.
- Designed **Content-Based Filtering (CBF)** layers using **cosine similarity** to rank high-dimensional product features.

Samsung Research Institute Noida, India · Sep 2020 – Sep 2021
Software Engineer (CL 2)

- Architected a **continuous on-device perception pipeline** ingesting multi-modal media streams for automated ML classification.
- Designed a **cross-profile data brokering engine** leveraging multi-user storage isolation to route assets from **User 1** to sandboxed **User 0**.

Teaching

University of California, Santa Barbara Santa Barbara, CA
Graduate Teaching Assistant Sep 2021 – Jun 2023

- Directed instructional labs, formulated algorithmic programming assignments, and monitored system architecture milestones for senior capstone teams.
- Covered advanced and core tracks: CS189 (Capstone), CS184 (Android), CS24 (C++ Data Structures), and CS16 (Problem Solving).

Technical Skills

Programming: Python, C++, Java, Kotlin, MATLAB

Robotics & Simulation: ROS/ROS2, Gazebo, RViz, Unity

Machine Learning & Vision: PyTorch, OpenCV, CUDA/GPU Acceleration

Methods: State Estimation (EKF), Model Predictive Control (MPC), Motion Planning, Sensor Fusion, Spatial Perception

Developer Tools: Git

Academic Service & Peer Review

Peer Reviewer, UIST 2024
ACM Symposium on User Interface Software and Technology

- Reviewed submissions on interactive systems, spatial tracking, and interface hardware.

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

STATEMENT OF GRADES OBTAINED FOR THE 8 SEMESTER COURSE IN ENGINEERING/TECHNOLOGY LEADING TO THE AWARD OF

BACHELOR OF TECHNOLOGY (HONOURS)

Roll No: 16EE10042
Name: SATYAM AWASTHI
Year of Admission : 2016-2017
Year of Graduation : 2019-2020
Course: B.Tech.(Hons.) in ELECTRICAL ENGINEERING

For Semester 1 SGPA: 8.09 CGPA: 8.09

Subno	Name	L-T-P	CRD	GRD
CY11001	CHEMISTRY	3-1-0	4	B
CY19001	CHEMISTRY LAB.	0-0-3	2	EX
EA10001	EXTRA ACADEMIC ACTIVITY-I	0-0-3	0	B
EE11001	ELECTRICAL TECHNOLOGY	3-1-0	4	C
EE19001	ELECTRICAL TECHNOLOGY LAB.	0-0-3	2	B
HS13001	ENGLISH FOR COMMUNICATION	3-0-2	4	B
MA10001	MATHEMATICS-I	3-1-0	4	B
ME19001	INTRODUCTION TO MANUFACTURING PROCESSES	0-0-3	2	A

For Semester 2 SGPA: 8.91 CGPA: 8.51

Subno	Name	L-T-P	CRD	GRD
CE13001	ENGINEERING DRAWING AND COMPUTER GRAPHICS	1-0-3	3	A
CS10001	PROGRAMMING AND DATA STRUCTURES	3-0-0	3	EX
CS19101	PROGRAMMING AND DATA STRUCTURES TUTORIAL AND LABORATORY	0-1-3	3	EX
EA10002	EXTRA ACADEMIC ACTIVITY-II	0-0-3	0	B
MA10002	MATHEMATICS-II	3-1-0	4	A
ME10001	MECHANICS	3-1-0	4	B
PH11001	PHYSICS	3-1-0	4	B
PH19001	PHYSICS LAB.	0-0-3	2	A

For Semester 3 SGPA: 8.68 CGPA: 8.57

Subno	Name	L-T-P	CRD	GRD
EA10003	EXTRA ACADEMIC ACTIVITY-III	0-0-3	0	Y
EC21103	INTRODUCTION TO ELECTRONICS	3-1-0	4	B
EC29003	INTRODUCTION TO ELECTRONICS LAB.	0-0-3	2	B
EE21101	SIGNALS & NETWORKS	3-1-0	4	B
EE29001	SIGNALS & NETWORKS LAB.	0-0-3	2	A
HS30103	ECONOMICS OF HEALTH	3-0-0	3	EX
MA20101	TRANSFORM CALCULUS	3-0-0	3	A
PH20003	PHYSICS-II	3-1-0	4	A

For Semester 4 SGPA: 8.96 CGPA: 8.67

Subno	Name	L-T-P	CRD	GRD
BS20001	SCIENCE OF LIVING SYSTEM	2-0-0	2	A
EA10004	EXTRA ACADEMIC ACTIVITY-IV	0-0-3	0	Y
EC21008	ANALOG ELECTRONIC CIRCUITS	3-1-0	4	A
EC29008	ANALOG CIRCUITS LAB.	0-0-3	2	EX
EE21002	ELECTRICAL MACHINES	3-1-0	4	A
EE21004	MEASUREMENTS AND ELECTRONIC INSTRUMENTS	3-1-0	4	A
EE29002	ELECTRICAL MACHINES LAB.	0-0-3	2	A
EE29004	MEASUREMENTS AND ELECTRONIC INSTRUMENTS LAB.	0-0-3	2	EX
EV20001	ENVIRONMENTAL SCIENCE	2-0-0	2	B
HS30074	SCIENCE AND HUMANISM	3-0-0	3	B

For Semester 5 SGPA: 9.04 CGPA: 8.75

Subno	Name	L-T-P	CRD	GRD
CS40019	IMAGE PROCESSING	3-0-0	3	A
EC31003(*)	DIGITAL ELECTRONIC CIRCUITS	3-1-0	4	A
EC39003(*)	DIGITAL ELECTRONIC CIRCUITS LAB.	0-0-3	2	EX
EE31009	CONTROL SYSTEM ENGINEERING	3-1-0	4	A
EE31011	POWER ELECTRONICS	3-1-0	4	B
EE39006	POWER ELECTRONICS LAB.	0-0-3	2	EX
EE39009	CONTROL AND INSTRUMENTATION LAB.	0-0-3	2	B
EP60025	SPECIAL TOPICS IN ENTREPRENEURSHIP	2-1-0	3	EX

For Semester 6 SGPA: 9.46 CGPA: 8.87

Subno	Name	L-T-P	CRD	GRD
CS31702(*)	COMPUTER ARCHITECTURE AND OPERATING SYSTEM	4-0-0	4	EX
CS60092	INFORMATION RETRIEVAL	3-0-0	3	B
EE30004	EMBEDDED SYSTEMS	3-0-0	3	EX
EE30012	ELECTROMAGNETIC THEORY & APPL	3-0-0	3	A
EE31002	POWER SYSTEMS	3-1-0	4	A
EE39002	POWER SYSTEMS LAB.	0-0-3	2	EX
EE39004	EMBEDDED SYSTEMS LAB.	0-0-3	2	EX
MA20106	PROBABILITY & STOCHASTIC PROCESSES	3-0-0	3	EX

For Semester 7 SGPA: 9.25 CGPA: 8.92

Subno	Name	L-T-P	CRD	GRD
CS60009(*)	SMARTPHONE COMPUTING & APPLICATION	3-0-0	3	EX
CS60021	SCALABLE DATA MINING	3-0-0	3	A
EE41001	INDUSTRIAL INSTRUMENTATION	3-1-0	4	B
EE47009	PROJECT-I	0-0-0	3	EX
EE48001	INDUSTRIAL TRAINING	0-0-0	2	A
EE49001	CONTROL AND ELECTRONIC SYSTEM DESIGN	0-0-3	2	A
EP60044	FRUGAL ENGINEERING	3-0-0	3	EX

For Semester 8 SGPA: 9.63 CGPA: 8.98

Subno	Name	L-T-P	CRD	GRD
CS31006	COMPUTER NETWORKS	3-0-0	3	EX
EE40002	ELECTRIC DRIVES	3-0-0	3	B
EE47008	PROJECT - PART 2	0-0-9	6	EX
EE48002	COMPREHENSIVE VIVA VOCE	0-0-0	2	EX
EE49004	POWER APPARATUS & SYSTEM DESIGN	0-0-3	2	EX

Additional subjects taken into account for earning a Minor

Subno	Name	L-T-P	CRD	Semno	GRD
CS21003	ALGORITHMS - I	3-1-0	4	4	A
CS29003	ALGORITHMS LABORATORY	0-0-3	2	4	A
CS60050	MACHINE LEARNING	3-0-0	3	5	B
CS60118	CLOUD COMPUTING	3-0-0	3	8	EX

* sign against a major curricular subject indicates that it has been taken into account for Minor

GPA in Minor: 9.36 Minor in : COMPUTER SCIENCE AND ENGINEERING

Details of other additional subjects

Subno	Name	L-T-P	CRD	Semno	GRD
EC21107	SEMICONDUCTOR DEVICES	3-1-0	4	3	A

Total Additional Credit Taken: 16 Total Additional Credit Cleared: 16
GPA in Additional Subjects: 9.00

Total Credit Taken in Major Curriculum: 176 Total Credit Cleared: 176 CGPA: 8.98

Joint Registrar (Academic)
IIT Kharagpur

Checked by

Joint Registrar (Academic) / Assistant Registrar(UG)

Date of Issue : 16 Jul 2020

University of California, Santa Barbara

Office of the Registrar, Santa Barbara, CA 93106-2015

PARCHMENT: TWCTPV1N

PRINTED: 04/05/23

OFFICIAL TRANSCRIPT

STUDENT NAME: SATYAM AWASTHI

PERM NUMBER: 807498

SSN: XXX-XX-4913
Birth Date: 01/23/XX

CURRENT ACADEMIC PROGRAM:

COLLEGE: College of Engineering
OBJECTIVE: Master of Science
MAJOR: Computer Science

COURSE	TITLE	GRADE	COMPLETE UNITS	GPA UNITS	GRADE POINTS	GPA	
2021 Fall Quarter							
	Admitted into the Master of Science Program in Computer Science						
	Admitted into the Master of Science Program in Computer Science						
CMPSC 291A	SPEC TOPIC APPS GEN	A+	4.0	4.0	16.00		
CMPSC 501	TECH COMP SCI TEACH	S	1.0	0.0	0.00		
CMPSC 502	TEACHING COMPTR SCI	S	3.0	0.0	0.00		
LING 4	INDIV & GRP INSTRUC	P	0.0	0.0	0.00		No Baccalaureate Credit Lower-Division Course
	Graduate Total:		8.0	4.0	16.00	4.00	
THRU F21	Graduate Total:		8.0	4.0	16.00	4.00	
2022 Winter Quarter							
CMPSC 271	ADV DIST SYS	A	4.0	4.0	16.00		
CMPSC 291A	SPEC TOPIC APPS GEN	A	4.0	4.0	16.00		
CMPSC 293C	SPEC TOP SYS PROG L	A	4.0	4.0	16.00		
	Graduate Total:		12.0	12.0	48.00	4.00	
THRU W22	Graduate Total:		20.0	16.0	64.00	4.00	
2022 Spring Quarter							
CMPSC 292C	SPEC TOP FOUN PROG	A+	4.0	4.0	16.00		
CMPSC 293N	SPEC TOPICS CS SYS	A	4.0	4.0	16.00		
CMPSC 595C	PRG LANG SOFT ENGR	S	2.0	0.0	0.00		
	Graduate Total:		10.0	8.0	32.00	4.00	
THRU S22	Graduate Total:		30.0	24.0	96.00	4.00	
2022 Summer Session							
CMPSC 193	INTERN IN INDUSTRY	P	1.0	0.0	0.00		
	Graduate Total:		1.0	0.0	0.00	0.00	
THRU M22	Graduate Total:		31.0	24.0	96.00	4.00	

CONTINUED TO PAGE 2

ISSUED TO:

SATYAM AWASTHI
6510 El Colegio Rd Apt 1102
Santa Barbara CA 93106-4200

Anthony Schmid
University Registrar

University of California, Santa Barbara
Office of the Registrar, Santa Barbara, CA 93106-2015

PARCHMENT: TWCTPV1N

PRINTED: 04/05/23

OFFICIAL TRANSCRIPT

STUDENT NAME: SATYAM AWASTHI PERM NUMBER: 807498

COURSE		TITLE	GRADE	COMPLETE UNITS	GPA UNITS	GRADE POINTS	GPA
2022 Fall Quarter							
CMPSC	263	RUNTIME SYSTEMS	A	4.0	4.0	16.00	
CMPSC	291A	SPEC TOPIC APPS GEN	B+	4.0	4.0	13.20	
CMPSC	596	DIRECTED RESEARCH	S	4.0	0.0	0.00	
		Graduate Total:		12.0	8.0	29.20	3.65
THRU F22		Graduate Total:		43.0	32.0	125.20	3.91
2023 Winter Quarter							
CMPSC	502	TEACHING COMPTR SCI	S	4.0	0.0	0.00	
CMPSC	596	DIRECTED RESEARCH	S	4.0	0.0	0.00	
		Graduate Total:		8.0	0.0	0.00	0.00
THRU W23		Graduate Total:		51.0	32.0	125.20	3.91
UC Credit		Graduate Total:		51.0	32.0	125.20	3.91

** END OF RECORD **

ISSUED TO:

SATYAM AWASTHI
6510 El Colegio Rd Apt 1102
Santa Barbara CA 93106-4200

Anthony Schmid
University Registrar

UNIVERSITY OF CALIFORNIA • SANTA BARBARA
TRANSCRIPT GUIDE

UNITS OF CREDIT:
Until September 1966, credits were recorded as semester units. Since that time, the University of California, Santa Barbara has operated on a quarter calendar, with three quarters per academic year and an optional summer session. Each quarter has ten weeks of instruction at the rate of one unit for every three hours of student work required each week.

CURRENT GRADING SYSTEM:
The grades A, B, C, and D may be modified by plus (+) or minus (-) suffixes. Grade points for each unit are assigned as follows:

A+ = 4.0	A = 4.0	A- = 3.7
B+ = 3.3	B = 3.0	B- = 2.7
C+ = 2.3	C = 2.0	C- = 1.7
D+ = 1.3	D = 1.0	D- = 0.7

Grades of F, I, IP, P, NP, S, U and W = 0.0 grade points

Unit credit, but not grade-point credit, is assigned for P and S grades.

P	=	Passed (C or better)	Undergraduate Course Only
NP	=	Not Passed (C- or below)	Undergraduate Course Only
S	=	Satisfactory (B or better)	Graduate Course Only
U	=	Unsatisfactory (B- or below)	Graduate Course Only
I	=	Incomplete	
W	=	Withdrawal after established deadline	
IP	=	Grade to be assigned at the end of the final course in the sequence	

GOOD ACADEMIC STANDING:
A student is in good standing academically unless otherwise indicated.

TRANSFER CREDIT:
Only credit that is accepted by the University is shown on the transcripts of UCSB students. Credit from other UC campuses may be shown either by individual course records or as a unit summary by campus. Individual courses from schools outside of the UC system are not shown.

ADVANCED PLACEMENT CREDIT:
Examinations and the credits accepted for them are shown on the transcript in the same format and location as transfer credit.

COURSE NUMBERING SYSTEM:

1 - 99	Lower Division Courses
100 - 199	Upper Division Courses
200 - 299	Graduate Courses
300 - 399	Professional courses for teachers
400 & above	Other professional and graduate courses

REPEATED COURSES:
An undergraduate student may repeat only those courses for which an NP or letter grade of C- or lower is recorded. If the grade from a previous attempt had been included in the GPA calculations, its effect is then removed by a "repeat adjustment" up to a maximum of 16 units over the student's career.

ACCREDITATION:
Western Association of Schools and Colleges (WASC)